SDG Blockchain Accelerator Cohort 2
Deliverables Roadmap

Accelerator Duration: 29 September - 15 December 2025
Challenge Owner (CO): UNDP El Salvador
Solution Maker (SM): Plastiks (by Nozama Tech Ltd.)

1. Challenge Definition

1.1 Refined Challenge Statement

El Salvador faces significant challenges in solid waste management, characterized by low recycling rates, lack of traceability, and the absence of an operational Extended Producer Responsibility (EPR) system. Plastic waste recovery relies heavily on informal grassroots recyclers who operate in precarious conditions without formal recognition or access to fair economic incentives. The challenge is to enable transparent, verifiable, and scalable systems that support plastic waste traceability, certification, and monetization while strengthening coordination across public, private, and community stakeholders.

1.2 Local Context

El Salvador generates approximately 4,200 tonnes of waste per day, of which only 5% is recycled. Plastics represent around 25% of municipal solid waste. Recovery systems remain informal, and recyclers lack recognition and economic security. There is currently no operational EPR framework.

1.3 Alignment with UNDP Country Office and National Development Priorities

Explain how the proposed solution aligns with the strategic objectives of the UNDP Country Office and national development frameworks. Does it support existing programs or government priorities (e.g., digital transformation, climate resilience, youth employment, financial inclusion)? Highlight any synergies with UNDP initiatives, government strategies, or donor-funded efforts.

*The proposed solution aligns strongly with UNDP El Salvador's strategic objectives and national development priorities by combining **digital public innovation** with **environmental governance and inclusive livelihoods** to strengthen the recycling ecosystem and prepare the country for effective **Extended Producer Responsibility (EPR)** implementation.*

12. Roadmap UNDPxPlastiks (1)

Alignment with UNDP Country Office strategic objectives

- ***Climate & environmental sustainability / circular economy:*** *The solution addresses low recycling rates and plastic leakage by enabling scalable recovery verification and incentivizing collection and recycling—supporting improved solid waste management outcomes and circular economy practices.*
  12. Roadmap UNDPxPlastiks (1)
- ***Governance, transparency, and accountability:*** *By producing **transparent, auditable data** on plastic recovery, the platform strengthens evidence-based decision-making and accountability across public and private stakeholders—key for future EPR compliance and environmental oversight.*
  12. Roadmap UNDPxPlastiks (1)
- ***Inclusive development and livelihoods:*** *The approach explicitly recognizes and economically empowers **grassroots recyclers**, improving inclusion in value chains and supporting more dignified income opportunities for a group currently working in precarious conditions.*
  12. Roadmap UNDPxPlastiks (1)
- ***Innovation and digital transformation:*** *The project introduces a **blockchain-based digital platform** for traceability, certification, and monetization—directly aligned with UNDP's push for responsible digital innovation to improve systems and service delivery.*
  12. Roadmap UNDPxPlastiks (1)

Alignment with national development frameworks and government priorities

- ***Solid waste management modernization:*** *The solution responds to a clear national systems gap—low recycling (5%) and limited traceability—by enabling interoperable, verifiable tracking of plastic recovery and ecosystem coordination.*
  12. Roadmap UNDPxPlastiks (1)

- ***EPR readiness and regulatory coordination:*** *El Salvador currently has **no operational EPR system**, and the solution supports the multi-stakeholder coordination required to operationalize EPR by providing the digital infrastructure and verification layer needed for compliance and reporting.*
  12. Roadmap UNDPxPlastiks (1)

Synergies with UNDP initiatives and donor-funded efforts

- ***Partnership-building and resource mobilization:*** *The roadmap prioritizes coalition-building with key actors (e.g., **FUNDEMAS, ANDRES, MARN**) and*

formalizing collaboration through partnership instruments (e.g., MoU), consistent with UNDP's convening role and partnership-based delivery model.
12. Roadmap UNDPxPlastiks (1)

- ***Donor alignment and blended financing potential:*** *The proposal is designed to be attractive to climate and innovation funding channels already identified in the roadmap—**Fund15 Project Catalyst, Small Grants Programme, Foundation Valéry, International Climate Finance**—creating a clear pathway to co-financing and scale.*
  12. Roadmap UNDPxPlastiks (1)
- ***Due diligence and risk management:*** *The inclusion of a robust due diligence process and compliance documentation supports UNDP's institutional requirements for responsible partnerships and implementation readiness.*
  12. Roadmap UNDPxPlastiks (1)

1.4 Relevance to UNDP Programmes and Resource Mapping
(Is the pilot building on an existing initiative, or is it a new standalone project? If the project builds on an existing initiative, provide further details. Include scope, scale, timeline and available co-financing or in-kind contributions.

*This initiative is designed as a **new pilot project** under the SDG Blockchain Accelerator, while remaining fully aligned with UNDP El Salvador's ongoing programme areas in **environmental sustainability, climate action, digital innovation, and inclusive economic development**. Although it does not directly build on an existing operational project, it is intentionally structured to **complement and strengthen UNDP's current and planned interventions** in circular economy promotion, and policy readiness for Extended Producer Responsibility (EPR), and mobilize resources for solid waste management initiatives.*

12. Roadmap UNDPxPlastiks (1)

*The pilot serves as a **foundational digital public good**, providing the traceability, certification, and data infrastructure required to support future UNDP-supported programmes related to environmental governance, private sector engagement, and climate finance mobilization. By generating transparent and auditable data on plastic waste recovery, the solution creates an enabling layer that can be integrated into broader UNDP initiatives on climate mitigation, sustainable cities, and inclusive livelihoods.*

12. Roadmap UNDPxPlastiks (1)

*From a **resource mapping and financing perspective**, the roadmap identifies multiple aligned funding pathways to support pilot implementation and subsequent scaling. These include:*

- ***Fund15 Project Catalyst**, aligned with blockchain-enabled climate and sustainability innovation;*
- ***Small Grants Programme (SGP)**, supporting community-based environmental action and grassroots empowerment;*

- ***Foundation Valéry**, aligned with innovation, sustainability, and social impact objectives;*
- ***International Climate Finance** mechanisms, leveraging verified impact data to attract results-based and blended finance.*
  12. Roadmap UNDPxPlastiks (1)

*In addition to external grant funding, the project is designed to catalyze **co-financing and in-kind contributions** from private sector partners and ecosystem actors engaged through FUNDEMAS, ANDRES, MARN, and sponsor companies. This blended approach reduces reliance on a single funding source and aligns with UNDP's emphasis on **leveraging partnerships and innovative financing models** to ensure sustainability beyond the pilot phase.*

1.5 Key Stakeholders & Potential Partners

(Please list all the partners involved in this project. List all relevant parties who will interact with, benefit from, or influence the solution (this may include government agencies, NGOs, community members, or tech partners.))

*The successful design, implementation, and scaling of the proposed pilot relies on a **multi-stakeholder ecosystem** that brings together public institutions, international organizations, private sector actors, civil society, and community-level participants. Key stakeholders and potential partners include:*

- ***UNDP El Salvador** – Acts as the **Challenge Owner and convening authority**, providing strategic oversight, governance, and coordination across stakeholders. UNDP ensures alignment with national development priorities, safeguards, and results-based management, while facilitating partnerships, resource mobilization, and donor engagement.*

- ***Plastiks (by Nozama Tech Ltd.)** – Serves as the **Solution Maker and technology partner**, responsible for the design, deployment, and maintenance of the blockchain-based platform. Plastiks provides the technical infrastructure for traceability, certification, and monetization of plastic waste recovery, as well as capacity-building support for users and partners.*

- ***MARN (Ministry of Environment and Natural Resources)** – Functions as a **key public sector stakeholder** ensuring policy alignment with national environmental regulations and waste management strategies. MARN's involvement supports regulatory coherence, institutional adoption, and future integration of the solution into EPR frameworks and national reporting systems.*

- ***FUNDEMAS (Fundación Empresarial para la Acción Social)** – Acts as a **strategic private sector and sustainability partner**, supporting engagement with producer companies, recyclers, and value-chain actors. FUNDEMAS facilitates partnerships,*

promotes corporate sustainability commitments, and plays a critical role in mobilizing private sector participation and co-financing.

- ***ANDRES (National Association of Solid Waste Management)*** *– Provides* ***sectoral expertise and operational insight*** *into waste management and recycling systems. ANDRES supports pilot implementation by connecting the project with waste management operators, recyclers, and technical networks across the country.*

- ***Grassroots recyclers and recycler organizations*** *– Are the* ***primary beneficiaries and end users*** *of the solution. They contribute operational data on plastic recovery, participate in testing and validation of the platform, and benefit from improved recognition, transparency, and access to fair economic incentives through verified recovery records.*

- ***Sponsor companies / producers*** *– Represent* ***current and future financing and demand-side partners****, particularly in the context of EPR readiness and corporate sustainability commitments. Their participation enables funding for the pilot, validation of compliance use cases, and long-term sustainability through Plastic Credit demand and reporting needs.*

Together, these stakeholders form an integrated ecosystem that supports ***shared ownership, coordinated implementation, and long-term scalability*** *of the solution, while reinforcing UNDP's role as a neutral broker and innovation enabler across sectors.*

1.6 Expected Impact

(Outline intended outcomes from the CO's perspective. What would a successful pilot enable - policy change, improved services, empowerment, systems improvement, or scaling innovation?)

The proposed pilot is expected to generate ***system-level, social, and environmental impact*** *by addressing critical gaps in plastic waste management, data transparency, and inclusion within El Salvador's recycling ecosystem. Key expected impacts include:*

- ***Verified and auditable plastic recovery data:***
  The solution will generate ***transparent, immutable, and auditable records*** *of plastic waste recovery using blockchain technology. This will significantly improve data reliability for public institutions, private sector actors, and development partners, enabling evidence-based decision-making, impact reporting, and accountability across the waste management value chain.*

- ***Empowerment and formal recognition of grassroots recyclers:***
  By digitally recording and certifying recovery activities, grassroots recyclers gain ***formal recognition for their work****, increased visibility within the ecosystem, and access to fairer economic opportunities. This contributes to improved livelihoods, dignity of work, and social inclusion for communities that are currently operating in*

informal and precarious conditions.

- ***Improved coordination across stakeholders:***
  *The platform serves as a **shared digital coordination layer** for government entities, recyclers, private sector partners, and supporting organizations. Standardized data and shared visibility reduce fragmentation, improve collaboration, and enable more efficient planning, monitoring, and resource allocation.*

- ***Readiness for Extended Producer Responsibility (EPR) implementation:***
  *The pilot directly supports **EPR preparedness** by providing the traceability, verification, and reporting infrastructure required for producer compliance. This positions El Salvador to more effectively design, operationalize, and enforce future EPR policies, while reducing implementation risks and administrative burdens.*

- ***Scalable and replicable impact:***
  *The solution is designed to be **modular and scalable**, allowing replication across additional municipalities, regions, and waste streams. Lessons learned from the pilot will inform national roll-out strategies and potential replication in other countries, amplifying impact beyond the initial implementation.*

*Collectively, these outcomes contribute to **stronger environmental governance, inclusive economic development, and long-term sustainability**, while reinforcing UNDP's objectives of fostering innovation, partnerships, and systems transformation.*

1.7 Target SDGs and Indicators
(List up to three specific SDG indicators your challenge and proposed solution will directly contribute to. Focus on indicators you will actively address, not indirect ones.)

The proposed solution directly contributes to the following **Sustainable Development Goals (SDGs)** and **specific indicators**, which will be actively addressed and measured during the pilot and post-acceleration phases:

- **SDG 12 – Responsible Consumption and Production**
  Indicator 12.5.1: *National recycling rate, tonnes of material recycled*
  The solution directly supports increased plastic recycling by enabling verified tracking, certification, and reporting of plastic waste recovery. By improving traceability and data reliability, the platform contributes to measurable increases in recycled plastic volumes and strengthens national reporting on recycling performance.

- **SDG 8 – Decent Work and Economic Growth**
  Indicator 8.3.1: *Proportion of informal employment in non-agriculture employment, by sector and sex*
  Through the formal recognition and digital recording of recovery activities, the project

supports the integration of **informal grassroots recyclers** into more formalized value chains. This contributes to improved working conditions, income visibility, and progress toward reducing informality within the waste management sector.

- **SDG 13 – Climate Action**
  Indicator 13.2.1: *Number of countries that have integrated mitigation, adaptation, impact reduction and early warning into national development plans*
  By generating auditable data on plastic recovery and supporting circular economy practices, the solution strengthens the evidence base required to integrate waste management and emissions reduction measures into climate-related planning and reporting frameworks.

These indicators are **directly linked to the core functionality of the solution**—traceability, verification, and monetization of plastic waste recovery—and provide a clear framework for measuring environmental, social, and systems-level impact in alignment with UNDP results frameworks.

2. User Mapping & Testing Strategy

2.1 Primary User Persona

Describe the main user or beneficiary of your solution.

The primary users and direct beneficiaries of the solution are **grassroots plastic recyclers and recycler organizations** operating within El Salvador's informal and semi-formal waste management ecosystem. These users typically include individual waste pickers, small recycler groups, cooperatives, and community-based collection initiatives that play a critical role in plastic recovery but remain largely invisible within formal systems.

Profile characteristics:

- Operate at the collection, sorting, and aggregation stages of the plastic value chain
- Work in precarious conditions with limited access to formal contracts, social protection, or stable income
- Have minimal access to digital tools, financing, or market information
- Are rarely recognized or compensated for the environmental value they generate

Needs and challenges:

- Lack of formal recognition and proof of recovered volumes
- Limited bargaining power and unstable income streams
- Absence of transparent mechanisms to demonstrate environmental impact
- Fragmented coordination with buyers, government entities, and private sector actors

Value provided by the solution:
The platform enables recyclers to **digitally record, verify, and certify** their plastic recovery activities, creating a trusted record of impact. This empowers users by:

- Increasing visibility and recognition of their work
- Enabling fairer and more transparent compensation mechanisms
- Facilitating inclusion in EPR schemes and corporate sustainability programmes
- Improving access to partnerships, support programmes, and future financial incentives

By centering grassroots recyclers as the primary user persona, the solution ensures that technological innovation directly translates into **inclusive economic opportunity, decent work, and systemic environmental impact**, in line with UNDP's people-centered development approach.

2.2 User Story

Use this format: "As a [user], I want to [goal], so that [value]."

As a grassroots recycler, often working informally and without official recognition,
I want my plastic recovery activities to be transparently and securely recorded and verified,
so that I can prove the environmental value of my work, receive fair and timely compensation, gain recognition from authorities and companies, and access new opportunities within formal recycling and EPR systems.

This user story reflects the core challenge faced by recyclers in El Salvador: despite being essential to plastic recovery, their contributions are largely invisible within formal reporting, financing, and policy frameworks. By addressing this need, the solution ensures that transparency and technology directly translate into **economic inclusion, decent work, and long-term livelihood security** for the primary beneficiaries.

2.3 Testing Strategy

How and when will user testing be conducted?

User testing will be conducted through a **phased, participatory approach** during the accelerator and early pilot implementation stages to ensure that the solution is **usable, relevant, and adapted to local operational realities**, particularly for grassroots recyclers with varying levels of digital literacy.

Timing and phases of testing:

- **Early-stage testing (design validation):** Initial testing sessions will be carried out with selected recycler groups and partner organizations to validate user flows, data inputs, and core assumptions around traceability, certification, and reporting.

- **Pilot-stage testing (live use):** Once the platform is configured for the local context, recyclers will use the solution in real recovery activities. This will allow testing of data submission processes, verification workflows, and usability under real operating conditions.

- **Post-testing refinement:** Feedback gathered during live pilot use will inform iterative improvements prior to finalizing the pilot-ready solution.

Testing methods:

- Guided onboarding sessions and hands-on demonstrations

- Observation of users interacting with the platform during daily recovery activities

- Structured feedback sessions with recyclers, aggregators, and partner organizations

- Validation of data accuracy and verification workflows with institutional stakeholders

Key focus areas:

- Ease of use and accessibility for non-technical users

- Accuracy and reliability of recovery data capture

- Clarity of incentives, outputs, and value for recyclers

- Operational feasibility for partners and institutions

This testing strategy ensures that user feedback is integrated continuously, reducing adoption risks and increasing the likelihood of **successful uptake, trust, and long-term sustainability** of the solution.

2.4 Feedback Loop Design

How will feedback from users and stakeholders be gathered and integrated?
The feedback loop is designed as a **continuous, structured, and multi-level process** to ensure that insights from users and stakeholders are systematically captured, analyzed, and translated into improvements throughout the accelerator and pilot phases.

Feedback collection mechanisms:

- **User-level feedback:** Grassroots recyclers and recycler organizations will provide ongoing feedback through facilitated workshops, onboarding sessions, and periodic check-ins. Simple feedback tools (e.g. guided discussions, short surveys, and verbal inputs captured by field coordinators) will be used to accommodate varying levels of literacy and digital access.

- **Operational feedback:** Aggregators and implementation partners will provide feedback on workflow efficiency, data quality, and operational feasibility through regular coordination meetings and structured reporting templates.

- **Institutional feedback:** Public sector stakeholders, UNDP teams, and private sector partners will review verification processes, dashboards, and reporting outputs to assess alignment with policy, compliance, and reporting requirements.

Integration and decision-making process:

- Feedback will be documented, categorized (e.g. usability, data accuracy, incentives, governance), and reviewed on a regular basis by the project implementation team.

- High-priority issues affecting usability, trust, or data integrity will be addressed through rapid iterations of the platform and operational guidelines.

- Broader strategic feedback will inform updates to governance arrangements, partnership roles, and pilot design.

Feedback-to-action cycle:

- Feedback is collected continuously during implementation

- Insights are reviewed in periodic internal and partner review sessions

- Agreed changes are implemented and tested
- Updates and rationale are communicated back to users and partners

This closed-loop approach ensures that users and stakeholders see their feedback reflected in tangible improvements, reinforcing **trust, ownership, and long-term adoption**, while maintaining the rigor required for institutional and donor engagement.

3. Solution Overview

3.1 Solution Summary

Briefly describe your proposed blockchain-based solution.

The proposed solution is a **blockchain-based digital platform** that enables the **traceability, verification, and monetization of plastic waste recovery** across El Salvador's recycling ecosystem. The platform securely records plastic recovery activities at the point of collection and aggregation, creating **immutable and auditable data** that can be trusted by public institutions, private sector actors, and development partners.

By leveraging blockchain technology, the solution provides a transparent certification layer that links recovered plastic to verified environmental impact. This enables the generation of **Plastic Credits** and other compliance- or impact-related records that can be used for **Extended Producer Responsibility (EPR) readiness, corporate sustainability reporting, and climate finance mechanisms**.

The platform is designed to be **inclusive and user-centered**, allowing grassroots recyclers to easily register recovery activities while ensuring that data meets institutional standards for accuracy and verification. At the ecosystem level, it acts as a shared digital infrastructure that improves coordination among recyclers, aggregators, government agencies, and sponsor companies, laying the foundation for **scalable circular economy systems** and long-term sustainability.

3.2 Core Features and Functionalities

What are the most important features of your solution?

The solution is built around a set of core features designed to ensure **transparency, usability, and institutional credibility**, while directly supporting inclusive participation and scalable impact:

- **Digital registration of plastic recovery activities**
  Grassroots recyclers and recycler organizations can record plastic collection and recovery events through a simple digital interface. Data captured includes material type, quantity, location, date, and responsible actor, ensuring consistent and standardized reporting across the ecosystem.

- **Blockchain-based traceability and immutable records**
  All verified recovery data is recorded on blockchain, creating tamper-proof and auditable records. This ensures trust, prevents double counting, and provides a reliable source of truth for stakeholders requiring high data integrity.

- **Verification and certification workflows**
  The platform supports structured verification processes involving aggregators and institutional stakeholders. Once validated, recovery data is certified, enabling its use for compliance, reporting, and impact claims.

- **Plastic Credit generation and monetization enablement**
  Verified recovery data can be linked to the issuance of Plastic Credits or equivalent environmental units, creating financial incentives for recyclers and enabling sponsor companies to support recovery through measurable impact mechanisms.

- **Stakeholder dashboards and reporting tools**
  Dedicated dashboards provide real-time visibility into recovery volumes, certified impact, and participation metrics for UNDP, government agencies, and private sector partners. Reporting outputs are aligned with policy, donor, and corporate sustainability requirements.

- **User access control and role-based permissions**
  The platform includes differentiated access for recyclers, aggregators, administrators, and institutional users, ensuring data security, accountability, and appropriate governance across the system.

- **Scalability and interoperability**
  The solution is designed to be modular and adaptable, allowing expansion to additional regions, waste streams, or integration with future EPR systems and national reporting platforms.

Together, these features ensure that the solution is **practically usable at the community level**, while meeting the **technical, governance, and reporting standards** required for long-term adoption and scaling.

3.3 Tech Stack and Tools

Which technologies and platforms will you use?

The solution will use a **hybrid architecture** combining a user-friendly digital platform with blockchain-based integrity and certification, ensuring it is practical for field use while meeting institutional requirements for transparency and auditability.

- **Web-based platform (user interface + admin dashboards)**
  A browser-accessible platform for recyclers, aggregators, and partner organizations to submit recovery data, manage verification steps, and view dashboards and reports.

- **Blockchain layer (traceability + immutable audit trail)**
  A blockchain network to record verified recovery events and certification outcomes, ensuring tamper-proof records, preventing double counting, and enabling transparent auditing.

- **Smart contracts / on-chain logic (where required)**
  Used to automate key blockchain interactions such as certification status recording and rules for credit issuance or proof-of-recovery statements.

- **Off-chain database and storage (operational data + documents)**
  A secure off-chain database for operational records (user accounts, workflow status, metadata) and document storage for supporting evidence, with hashes or references anchored on-chain where appropriate.

- **Data analytics and reporting tools**
  Built-in reporting dashboards and exportable reports for UNDP, government partners, and sponsor companies to support monitoring, compliance, and donor/corporate reporting.

- **User access management and security**
  Role-based access control, authentication, and audit logs to ensure appropriate permissions and accountability across recyclers, aggregators, and institutional stakeholders.

This stack ensures a balance between **field usability**, **scalability**, and **institutional-grade transparency**, while keeping the solution flexible for future integration into EPR systems and national reporting frameworks.

3.4 Blockchain Functionality

What blockchain-specific interactions are included (e.g., token minting, smart contracts, on-chain verification)?

The blockchain component is used as a **trust and integrity layer** to ensure that plastic recovery data and certification outcomes are **tamper-proof, auditable, and credible** for government, UNDP, donors, and private sector partners. Key blockchain-specific interactions include:

- **On-chain recording of verified recovery events**
  Once plastic recovery data is submitted and validated through the verification workflow, a corresponding record (or cryptographic proof) is written to the blockchain. This creates an immutable audit trail that prevents manipulation and supports transparent reporting.

- **On-chain certification / proof-of-recovery status**
  The platform records certification outcomes on-chain (e.g., “verified,” “certified,” “rejected”), ensuring that verification decisions are traceable and can be independently audited by authorized stakeholders.

- **Smart contract–based verification logic (as applicable)**
  Smart contracts can be used to enforce standardized rules for recording verification results, timestamps, and actor roles (e.g., recycler → aggregator → verifier), ensuring consistent application of certification criteria and reducing disputes.

- **Prevention of double counting and fraud risk reduction**
  By anchoring verified recovery records on-chain, the solution reduces the risk of duplicated claims and supports integrity checks across actors and locations—critical for EPR compliance and credible impact finance.

- **Plastic Credit enablement (issuance linkage)**
  Certified recovery records can be linked to the generation/issuance of **Plastic Credits** (or equivalent environmental units). Where credit issuance is implemented, blockchain records provide the provenance and traceability required to demonstrate that credits correspond to verified recovery.

- **Auditability for institutional and donor reporting**
  Blockchain provides an independent, time-stamped record that enables UNDP and partners to demonstrate transparency, strengthen accountability, and support results-based financing or donor assurance requirements.

These blockchain interactions ensure the solution delivers **verifiable traceability and certification**, enabling credible monetization mechanisms and long-term readiness for EPR implementation.

4. Weekly Workplan

Briefly describe your team's focus during each stage of the program. (You may use the workplan template from your hackathon roadmap.)

The team's workplan follows the **pilot roadmap and milestones presented in Slide 8** of the UNDP–Plastiks Pilot Demo Day presentation, translating them into accelerator-aligned phases and weekly focus areas.

Weeks 1–4 | Milestone 1 – Design & Localization
Focus: Solution design, localization, and institutional alignment

- Scope and define the accreditation and verification framework
- Develop standardized recovery and certification workflows
- Localize the platform to **Spanish language** and national regulatory requirements
- Address data protection and privacy compliance
- Initiate close collaboration with **FUNDEMAS** and key institutional stakeholders
- Align solution design with MARN and ANDRES policy frameworks

Weeks 5–8 | Milestone 2 – Accreditation & Onboarding
Focus: Ecosystem onboarding and operational readiness

- Accredit **2–3 waste management actors** (aggregators / facilities)
- Onboard **150–200 grassroots waste pickers** onto the platform
- Deliver hands-on training and digital onboarding sessions
- Connect the recovery data repository to reporting and monitoring systems
- Test verification workflows and refine operational processes

Weeks 9–11 | Milestone 3 – Tokenization & Marketplace Launch
Focus: Impact monetization and market activation

- Tokenize and certify **300–400 tons of recovered plastic**
- Issue the first **Plastic Credits** linked to verified recovery
- Launch the corporate-facing marketplace for sponsor companies
- Validate traceability, credit issuance, and auditability mechanisms
- Engage private sector partners on sustainability and EPR use cases

Weeks 12–13 | Milestone 4 – Reporting & Scale-Up Planning
Focus: Results, visibility, and future scaling

- Publish an **impact dashboard** with verified recovery and participation data
- Host a multi-stakeholder workshop to present results and lessons learned
- Develop a national and regional **scale-up and replication plan**
- Define a tokenization roadmap and revenue framework
- Consolidate inputs for donor reporting, fundraising, and post-acceleration planning

This phased workplan ensures a **logical progression from design to deployment, monetization, and scale**, while maintaining continuous stakeholder engagement and alignment with UNDP objectives and national priorities.

5. Final Deliverables

5.1 Solution Demo (PoC)

A working or visual prototype that demonstrates the core idea. (To be scheduled and presented virtually to CO team and UNDP/Partner FPs in a weekly coordination meeting)

5.2 Testing & Feedback Documentation

Summary of user testing activities, challenges, and feedback.

5.3 Feedback Integration Summary

Description of what changes were made based on testing and feedback.

5.4 Pilot-Ready Solution

A stable and functional version of the solution, ready for pilot deployment.

5.5 Confirmed Pilot Partnership(s)

Documented confirmation of at least one partner committed to implementation.

5.6 Stakeholder Alignment Matrix

Defined roles, responsibilities, and coordination plan with key partners

Stakeholder / Partner	Role in the Pilot	Key Responsibilities	Level of Involvement (Lead / Support / Advisory)	Engagement Modality (e.g. MoU, Contract)	Focal Point / Contact Person

UNDP El Salvador	Challenge Owner & Convenor	Strategic oversight; governance and safeguards; stakeholder coordination; alignment with national priorities; donor engagement and reporting	Lead	MOU / Accelerator Framework	UNDP Country Office – Social Investment Team
Plastiks (by Nozama Tech Ltd.)	Solution Maker & Technology Provider	Platform design and localization; blockchain traceability and certification; technical support; data integrity and reporting	Lead	Contract / Partnership Agreement	Plastiks Project Lead
MARN (Ministry of Environment and Natural Resources)	Public Sector & Policy Partner	Policy alignment; regulatory guidance; integration with national waste management and EPR frameworks; validation of reporting relevance	Advisory	Letter of Support / Institutional Collaboration	MARN Environmental Management Unit
ANDRES (National Solid Waste Authority)	Sectoral & Operational Partner	Coordination with waste operators; operational insights; facilitation of pilot implementation	Support	Letter of Support / Institutional Collaboration	ANDRES Technical Directorate

FUNDEMAS	Private Sector & Sustainability Partner	Engagement of sponsor companies; private sector mobilization; support to partnerships and resource mobilization	Support	Letter of Support / Memorandum of Understanding (MoU)	FUNDEMAS Sustainability Program Lead
Waste Aggregators / Facilities	Implementation & Verification Partners	Accreditation participation; verification of recovery data; operational supervision; reporting inputs	Support	Participation Agreement	Facility Manager / Operations Lead
Grassroots Recyclers / Recycler Organizations	Primary Users & Beneficiaries	Plastic collection and recovery; data submission; participation in training, testing, and feedback activities	Lead (Operational)	Participation Agreement / Onboarding Consent	Recycler Cooperative Representative
Sponsor Companies / Producers	Financing & Demand-Side Partners	Pilot co-financing; support or purchase of Plastic Credits; use of verified data for sustainability and EPR reporting	Support	Sponsorship Agreement / Letter of Commitment	Corporate Sustainability Mana

5.7 Budget Proposal (Pilot)

Provide a budget breakdown for pilot implementation. Include estimated cost, unit, quantity, and funding source.

Budget Line Item	Unit / Description	Quantity	Unit Cost (EUR)	Total Cost (EUR)	Funding Source
One-Time Accreditation – Waste Recovery Aggregator	Accreditation fee	1	2,500	2,500	Pending
One-Time Accreditation – Recycling Entity	Accreditation fee	1	2,500	2,500	Pending
One-Time Accreditation – Waste Picker Profiling	200 waste pickers (€12.5 per picker)	200	12.5	2,500	Pending
Annual License & Verification – Waste Recovery Aggregator	Annual license & verification	1	5,000	5,000	Pending
Annual License & Verification – Recycling Entity	Annual license & verification	1	5,000	5,000	Pending
Annual License & Verification – Waste Picker Profiling	200 waste pickers (€25 per picker/year)	200	25	5,000	Pending

Project Management Fee	Pilot coordination & reporting	1	3,000	3,000	Pending

TOTAL ESTIMATED PILOT COST: €25,500

Revenue Sharing and Commercial Terms

Plastic Credit Verification: 20% of proceeds from Plastic Credit sales

Carbon Credit Verification: USD 2 per metric ton

Platform Access for UNDP: Free

5.8 Impact Measurement Framework

5.8.1 Contribution to relevant SDGs and indicators

The impact measurement framework is designed to ensure that the pilot generates ***measurable, verifiable, and policy-relevant outcomes****, directly aligned with selected* ***SDGs and UNDP results frameworks****. The use of blockchain-based traceability enables reliable data collection and auditability, strengthening the credibility of reported results.*

- ***SDG 12 – Responsible Consumption and Production***
  Indicator 12.5.1: National recycling rate, tonnes of material recycled
  The solution directly contributes to this indicator by generating ***verified, real-time data on plastic waste recovery and recycling volumes****. Certified recovery records provide an evidence base to measure increases in recycled plastic, improve national reporting, and support circular economy targets.*

- ***SDG 8 – Decent Work and Economic Growth***
  Indicator 8.3.1: Proportion of informal employment in non-agriculture employment, by sector and sex
  By formally recording and certifying the activities of grassroots recyclers, the project supports the ***progressive formalization of informal work*** *in the waste sector. Data on participation, income-related proxies, and verified recovery activities contribute to monitoring improvements in inclusion and economic opportunity.*

- ***SDG 13 – Climate Action***
  Indicator 13.2.1: Number of countries that have integrated climate mitigation measures into national development planning
  The platform provides ***auditable data on plastic recovery and associated emissions avoidance****, supporting the integration of waste management and circular economy actions into climate planning, reporting, and potential climate finance*

mechanisms.

*Through these indicators, the pilot links **on-the-ground recovery activities** to **national and international reporting frameworks**, ensuring that impact is not only achieved but also credibly measured and communicated to policymakers, donors, and private sector partners.*

5.8.2 Key Performance Indicators (KPIs)

KPI	*Measurement Method*	*Target / Threshold*	*Notes*
Tonnes of plastic recovered and verified	*Blockchain-certified recovery records aggregated through the platform*	*300–400 tonnes during pilot*	*Directly linked to SDG 12.5.1*
Number of grassroots recyclers onboarded	*Platform user registrations and verified profiles*	*150–200 recyclers*	*Measures inclusivity and adoption*
Number of accredited waste actors	*Accreditation records (aggregators / recycling entities)*	*2–3 accredited actors*	*Ensures operational capacity*
Percentage of recovery records verified	*Verification workflow completion rate*	*≥90%*	*Indicator of data quality*
Plastic Credits issued	*On-chain certification and issuance records*	*First issuance completed*	*Demonstrates monetization readiness*

Stakeholder engagement level	*Workshop attendance and coordination meeting records*	*≥3 multi-stakeholder sessions*	*Measures ecosystem alignment*
Platform adoption rate	*Usage analytics (active users vs onboarded users)*	*≥80% active users*	*Proxy for usability*
Impact reporting readiness	*Availability of dashboards and standardized reports*	*Dashboard published by end of pilot*	*Supports donor and policy reporting*

5.8.3 Monitoring & Reporting Plan

a) *Frequency:*
b) *Responsible Parties:*
c) *Format:*
d) *Tools:*

Frequency:

- *Ongoing / real-time: Continuous data capture of plastic recovery activities through the platform*
- *Monthly: Internal progress monitoring and operational reviews*
- *Quarterly (or end of pilot): Consolidated impact reporting for UNDP and partners*
- *End of pilot: Final impact, lessons learned, and scale-up report*

Responsible Parties:

- *Plastiks: Day-to-day monitoring of platform usage, data quality, verification workflows, and dashboard maintenance*
- *UNDP El Salvador: Oversight of results reporting, alignment with SDG indicators, and donor reporting requirements*
- *Aggregators / Accredited entities: Validation of recovery data and operational reporting inputs*

- *Institutional partners (e.g. MARN, ANDRES): Advisory review of reports for policy relevance*

Format:

- *Real-time digital dashboards*
- *Standardized progress and impact reports (tables, charts, narrative summaries)*
- *Final pilot report including KPIs, SDG contributions, risks, and lessons learned*

Tools:

- *Plastiks blockchain-based platform and dashboards*
- *On-chain verification and audit logs*
- *Off-chain reporting and data analytics tools*
- *Standard UNDP reporting templates (as applicable)*

5.8.4 Risks Assessment & Mitigation Plan

Risks	***Mitigation***
Low adoption of digital tools by grassroots recyclers	*Provide hands-on training, guided onboarding sessions, and continuous technical support; use simple, user-friendly interfaces adapted to local contexts*
Limited digital literacy or access to devices/connectivity	*Offer assisted data entry through aggregators; allow shared device use; adapt workflows to low-connectivity environments*

Data quality or inaccurate reporting	*Implement multi-step verification workflows; use standardized data collection protocols; apply blockchain-based immutability to prevent manipulation*
Risk of double counting or fraudulent claims	*Anchor verified recovery records on blockchain; enforce unique identifiers and verification rules; enable audit trails*
Limited coordination among institutional stakeholders	*Establish early stakeholder engagement; define clear roles and governance structures; use regular coordination meetings facilitated by UNDP*
Regulatory or policy changes affecting waste management or EPR	*Maintain continuous alignment with MARN and ANDRES; adapt workflows and reporting to evolving regulations*
Financial sustainability post-pilot	*Develop monetization mechanisms (Plastic Credits, verification fees); engage sponsor companies early; pursue blended financing*
Delays in funding or resource mobilization	*Identify multiple funding sources; phase implementation to match available resources; leverage in-kind contributions*
Reputational risk linked to environmental claims	*Ensure robust verification, transparent reporting, and independent auditability of data*

5.9 Post-Acceleration Roadmap & Sustainability Plan

5.9.1 Phased Implementation Timeline / Milestones

Phase	Timeframe	Key Milestones & Activities
Short Term	**0–6 months**	Design & Localization (Month 1): Scope accreditation and verification framework; define standardized workflows; localize platform to Spanish and national regulations; ensure data privacy compliance; establish collaboration with FUNDEMAS. Accreditation & Onboarding (Months 2–3): Accredit 2–3 waste actors; onboard 150–200 grassroots waste pickers; deliver training and digital onboarding; connect data repository to reporting systems.
Medium Term	**6–18 months**	Tokenization & Marketplace Launch (Months 4–6): Tokenize and certify 300–400 tons of recovered plastic; issue first Plastic Credits; launch corporate marketplace; validate traceability, auditability, and monetization mechanisms. Operational Consolidation: Strengthen partnerships; refine verification and governance processes; support early EPR and corporate sustainability use cases.
Long Term	**18+ months**	Reporting & Scale-Up: Publish impact dashboards and reports; host multi-stakeholder workshop; develop national and regional expansion plan; define long-term tokenization roadmap and revenue framework. System Integration: Support integration with future EPR systems; replicate model across regions and waste streams; leverage private co-financing and green finance instruments.

5.9.2 Budget for Phased Implementation (post-acceleration)

5.9.3 Financial Sustainability Model

How will the solution generate income post-pilot? Pricing model (e.g. subscriptions, licensing, transaction fees) and when it kicks in.

5.9.4 Partnership and Co-Financing Commitments

Which partners will be supporting the project (technical, financial, institutional) and what would be their contributions?

5.9.5 Fundraising Plan

Plan for securing funding through grants, ecosystem funds, investors or other sources. Targeted sources and fundraising timeline.

5.11 Legal & Compliance Considerations

Summary of key legal, regulatory, or compliance considerations.

5.12 Visibility Assets / Communication Materials

1–2 slides, visuals, or storylines for use by UNDP or partners to promote your solution.

5.13 Pitch Deck

Concise and compelling presentation for stakeholders, funders, and partners. ((Please upload to your google drive folder)

5.14 Final Demo Presentation

An engaging walkthrough of the live solution and its implementation roadmap at the end of the program.

Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

Deliverable	Target Completion (Week # or Date)
1. Challenge Definition (incl. local context, SDG contributions, partners, impact)	
2. User Mapping & Testing Strategy (personas, user stories, feedback loop)	

3. Solution Overview (core features, tech stack, blockchain functionality)	
5.1 Solution Demo (PoC)	
5.2 Testing & Feedback Documentation	
5.3 Feedback Integration Summary	
5.4 Pilot-Ready Solution	
5.5 Confirmed Pilot Partnership(s)	
5.6 Stakeholder Alignment Matrix	
5.7 Budget Proposal (Pilot)	
5.8 Impact Measurement Framework (KPIs, SDG indicators, risk plan)	
5.9 Sustainability Plan & Post-Acceleration Roadmap (implementation timeline, budget, partners, fundraising)	
5.11 Legal & Compliance Mapping	
5.12 Visibility Assets / Communication Materials	
5.13 Pitch Deck	